

The Libra Dialogue Protocol for Comparative Deliberation

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Abstract. Deliberative dialogue often requires agents to compare options publicly as relevant alternatives shift over the course of interaction. This paper introduces Libra, a dialogue protocol for comparative deliberation inspired by the Fatio protocol and informed by Tucker’s distinction between justifying and requiring weight of reasons. Libra preserves Fatio’s five locutions but reworks the content layer around option claims, contrastive reasons, explicit weight assignments, and grounds. The central technical contribution is a public semantics, formulated with deterministic structural equation models (SEMs), in which agents incrementally disclose SEM fragments rather than merely exchange abstract reasons. On this basis, the paper defines update rules for option introduction, changing comparison spaces, reason-element discussion, and retraction. A case study illustrates the framework.

Keywords. formal dialogue, weighing reasons, deontic reasoning, causal models, multi-agent systems

1. Introduction

Deliberative dialogue is often comparative: an agent publicly puts forward an option and may then be asked why that option should stand against the alternatives currently in play. Existing dialectical and argumentation-based models already provide rich tools for representing proposals, challenges, attacks, replies, and argument evaluation. Here we address the problem of devising a protocol in which the dialogue explicitly records which options are live, which rivals are relevant for each option, which grounds and weight assignments have been disclosed, and which justificatory burdens remain open as the comparison space changes.

This paper introduces Libra, a dialogue protocol for comparative deliberation inspired by the Fatio dialogue protocol [7,8] and informed by Tucker’s distinction between justifying and requiring weight [19]. Libra preserves Fatio’s five locutions, i.e., assertion, question, challenge, justification, and retraction, but reworks the content layer around options, contrastive reasons, and explicit weights. Assertions publicly advance options, questions request support for an asserted option against its relevant rivals, and challenges focus on a particular rival.

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We use deterministic structural equation models (SEMs) [3,15] to model that the dialogue records not only that a reason was offered, but also how publicly disclosed grounds and weights determine a comparative claim. SEMs provide a compact representation of explanatory dependency, keep the protocol layer separate from the explanatory layer, and make incremental update and retraction precise. The resulting architecture combines a public transition system with a public SEM layer: dialogue moves manage public commitments and burdens, while justifications extend a speaker's public SEM fragment with equations for pairwise permissibility claims.

Section 2 reviews the relevant background on Fatio, Tucker's account of reasons, and deterministic structural equation models. Section 3 presents the syntax of Libra. Section 4 gives the public SEM semantics. Section 5 develops the case study. Section 6 discusses related work. Section 7 concludes and outlines further research.

2. Background

2.1. Fatio

The language *FIPA ACL* for agents communications acts as the foundation on which Fatio was built upon [8]. The extension adds a proper theory of argumentation to the original language by introducing a set of locutions for requesting and providing reasons. The core idea is that agents are either proponents or opponents of claims aiming to arrive at truth by a sequence of dialectical moves.

The protocol consists of dialectical moves (illocutions) of which each has either the form *illocution*(P_i, ϕ) or *illocution*(P_i, P_j, ϕ) where P_i is the participant making the speech act (move), P_j is the target of the speech act and ϕ is the content. Any illocution is public to every participant and the syntax of the content can be any formal language. The set of all well-formed content formula is denoted $\mathcal{C}$. In some utterances the content contains arguments, where the set of well-formed argument formulae is denoted as $\mathcal{A}$. Given $\phi \in \mathcal{C}$ and $\Phi \in \mathcal{A}$ then $\Phi \vdash^+ \phi$ expresses that Φ is an argument in support of ϕ . Similarly $\Phi \vdash^- \phi$ then expresses Φ is in opposition of ϕ . There is a total of 5 illocutions [8].

- F1 *assert*(P_i, ϕ) : Participant P_i asserts the statement ϕ and creates a dialectical obligation for itself to provide justification for ϕ if requested by another participant.
- F2 *question*(P_j, P_i, ϕ) : Participant P_j requests a justification of a prior utterance of the form *assert*(P_i, ϕ) from participant P_i . There are no dialectical obligations on the part of participant P_j .
- F3 *challenge*(P_j, P_i, ϕ) : Participant P_j challenges a prior *assert*(P_i, ϕ) by a different participant P_i , thereby requesting a justification for ϕ . This differs from *question*(P_j, P_i, ϕ) by also introducing a dialectical obligation on P_j to defend not asserting ϕ .
- F4 *justify*($P_i, \Phi \vdash^+ \phi$) : Participant P_i that previously performed the dialectical move *assert*(P_i, ϕ) and was subsequently challenged or questioned is obligated to defend the statement ϕ . The set Φ should then provide the necessary arguments to justify ϕ . Similarly *justify*($P_i, \Phi \vdash^- \phi$).
- F5 *retract*(P_i, ϕ) : Participant P_i that previously stated *assert*(P_i, ϕ) retracts the previous statement. The dialectical obligation on P_i to defend ϕ is thereby withdrawn. Similarly a previous *justify*($P_i, \Phi \vdash^+ \phi$) can also be retracted.

To keep track of the obligations each participant has committed to they each possess a *dialectical obligations store* $DOS(P_i)$. These stores consists of triples (P_i, X, Y) where P_i is the participant, X is a well-formed formula in the content language and $Y \in \{+, -\}$. Given $(P_i, \phi, +) \in DOS(P_i)$, P_i has the obligation to justify the proposition ϕ in the positive, while $(P_i, \phi, -) \in DOS(P_i)$ would require justifying against ϕ . The semantics of each illocution is defined by a set of pre-conditions defining what is required to hold to perform the utterance, and post-conditions defining what changes in the setting after the utterance. In order to be specific two modal operators are introduced, $\{B_i, D_i\}$ for *belief* and *desire* of participant P_i . The full definitions can be found in [8].

We use Fatio as the starting point for two reasons. First, its five locutions are minimal and generic enough to be reinterpreted over a new content layer: options, contrastive reasons, and weight assignments. Second, its semantics is already organized around preconditions, postconditions, and public dialectical obligations, which matches our own update-based treatment of public SEM fragments.

2.2. Reasons

In this paper, we use key assumptions from Tucker’s theory [19] needed to understand the Libra protocol. For analysis of benchmark examples and details on philosophical concepts, motivations, and broader literature on reasons, see Tucker’s analysis.

Tournament Metaphor. Tucker’s use of “tournament” can be seen as a metaphor, because it uses the analogy between comparing options and comparing players in a competition: “ ϕ is permissible by winning a tournament, a pairwise competition with each alternative.” It suggests so-called *comparativism*: “reasons are always differences or comparisons between ϕ and some specified alternative” and so-called *contrastivism*: “for some possible ϕ , the reason for ϕ can vary as you vary the alternative.” Tucker discusses the following benchmark example called the Café or Kid Case Problem [5]. “There is a child stuck in a burning building. Your options are Bystander (hang around and watch the events unfold), Save1 (save the child and get severely burned in the process), and Café (keep your promise to meet your friend at the café).” This is modeled as a tournament in which Café wins the competition against Bystander, while the other two competitions are undecided. Reasons may be comparative or contrastive; for example, “the child’s death is a reason against Café in the pairwise competition with Save1 but not in the pairwise competition with Bystander”.

Dual Scale Metaphor. Dual scale. For Libra, Tucker’s account supplies the distinction between grounds and two roles of weight [19]. In the *Café-or-Kid* case [5], the relevant options are *Save1*, *Bystander*, and *Café*. A ground is a consideration that can bear weight for or against an option in a given comparison; in our running example, the child’s being in danger is represented by the ground g_{danger} . Tucker distinguishes two functional roles of reasons: justifying weight and requiring weight. Justifying weight measures how strongly a reason pushes an option toward permissibility, while requiring weight measures how strongly it pushes the relevant alternative toward impermissibility [19]. Libra abstracts from Tucker’s full aggregation theory and records these roles as public weight atoms $J(r, a, b) = w$ and $R(r, a, b) = w$, indexed to ordered pairs of options. This indexing is essential: the same ground can have different force when an option is compared with different rivals.

2.3. Causal models

A causal model represents a system by variables together with equations specifying how the values of some variables depend on the values of others. For the purposes of this paper, we work with Halpern and Pearl's framework of deterministic structural equation models together with interventions [15,3]. Our presentation is adapted from [4].

A deterministic structural equation model consists of a set U of exogenous variables, a set V of endogenous variables, a range assignment R that gives each variable $X \in U \cup V$ a set $R(X)$ of possible values, and for each endogenous variable $X \in V$ a structural equation $X := F_X(\text{PA}_X)$, where $\text{PA}_X \subseteq U \cup V \setminus \{X\}$ is the set of variables on which X immediately depends, and F_X maps admissible values of those parent variables to a value in $R(X)$. The exogenous variables are not determined by equations inside the model; intuitively, their values are fixed from outside the system. An assignment of values to the exogenous variables is called a *context*. Relative to a context u , the structural equations determine the values of the endogenous variables. When the model is acyclic, that is, when the endogenous variables can be ordered so that no variable depends on a later one in the order, every context induces a unique solution. This is the setting used in this paper.

An *intervention* on an endogenous variable X that sets it to a value $x \in R(X)$ is represented by replacing the structural equation for X with the constant equation $X := x$. More generally, one can intervene on a finite set of endogenous variables by simultaneously replacing each of their equations by constant assignments.

3. The syntax of Libra

This section fixes the public vocabulary of the dialogue. The basic idea is to separate the two layers of option claims and reasons with their weights.

Definition 3.1 (Locution language). Let Ag be the set of agents and O the set of options. The locution language is $m ::= \text{assert}(i, a) \mid \text{question}(j, i, x) \mid \text{challenge}(j, i, a, b) \mid \text{justify}(i, \Gamma) \mid \text{retract}(i, u)$, where $i, j \in \text{Ag}$, $a, b \in O$, x is an admissible question target, u is an admissible retraction target, and Γ is a justification package.

Assertions and challenges put options into play, questions and challenges open justificatory burdens, and justifications answer those burdens with finite public packages. Retractions withdraw previously committed options.

Definition 3.2 (Public atoms and weight scale). Let $G = G^{\text{exo}} \uplus G^{\text{qry}}$ be the set of ground atoms, where G^{exo} contains publicly fixed grounds and G^{qry} contains grounds that may still be discussed. Let $\mathcal{W} = \{\text{low}, \text{mid}, \text{high}\}$, $\text{low} \prec \text{mid} \prec \text{high}$. Let $\preceq$ be the reflexive closure of $\prec$. The atomic public contents are:

$\text{Opt}(a)$	option claim,
g	ground atom, where $g \in G$,
$\text{J}(r, a, b) = w$	justifying-weight atom,
$\text{R}(r, a, b) = w$	requiring-weight atom,

where $a, b \in O$, r is a reason label, and $w \in \mathcal{W}$. Let $W = \{J(r, a, b) = w, R(r, a, b) = w \mid a, b \in O, a \neq b, w \in \mathcal{W}\}$ be the set of public weight atoms.

Definition 3.3 (Public tokens). Let $\text{Tok} = \{\text{Opt}(a) \mid a \in O\} \uplus G^{\text{qry}} \uplus W$.

The point of Tok is to identify exactly those public items that can be owned, questioned, and later retracted during the dialogue. Thus exogenous grounds are neither question targets nor retraction targets: they are already publicly fixed.

Definition 3.4 (Contrastive reason elements). A contrastive reason element for a against b is a tuple $\rho = \langle g, J(r, a, b) = w_J, R(r, a, b) = w_R \rangle$, where $g \in G$, r is a reason label, and $w_J, w_R \in \mathcal{W}$. Its element set is $\text{elts}(\rho) = \{g, J(r, a, b) = w_J, R(r, a, b) = w_R\}$.

The content language is contrastive: grounds and weight assignments are indexed to ordered pairs of options, so that same reason may behave differently in different comparisons. A single contrastive reason element records one publicly stated comparison. It does not yet settle whether the overall case for a against b succeeds; that assessment is made at the package level in the semantics.

Package adequacy and justification packages. Whether a package of reason elements is adequate, and whether it answers a given outstanding burden, depends on semantic notions such as relevance sets and target types, and is therefore defined in Section 4.

Protocol scope. Libra preserves Fatio's five locutions, but it does not adopt Fatio's original combination rules. Instead, questions and challenges open outstanding justificatory targets that may be discharged by a later justification move.

4. A public SEM semantics

In this section we define the semantics of Libra by using *causal models*, and in particular deterministic SEMs with interventions [15,3]. The semantics separates dialogue bookkeeping from explanatory content. The public state records which options and reason-elements are in play, which justificatory burdens are outstanding, and which rivals are currently relevant. In parallel, each agent carries a public SEM fragment containing only the variables and equations that have actually been disclosed in the dialogue. Assertions and challenges create provisional permissibility commitments by intervention, justifications replace or extend those commitments with structural equations, and retractions use interventions to withdraw options or reason elements.

Definition 4.1 (Public state). The tuple $s_t = \langle \Omega_t, \text{Own}_t, \text{Rel}_t, \mathcal{O}_t, (\widehat{M}_i^t)_{i \in \text{Ag}} \rangle$ represents a public state at step t , where:

- $\Omega_t \subseteq O$ is the live option set;
- $\text{Own}_t : \text{Tok} \rightarrow 2^{\text{Ag}}$ is the public ownership map;
- $\text{Rel}_t(i, a) \subseteq O \setminus \{a\}$ is the public relevance set for agent i and option a ;
- $\mathcal{O}_t$ is the set of outstanding justificatory targets;
- $\widehat{M}_i^t$ is the current public SEM fragment attributed to agent i .

For an option token $\text{Opt}(a)$, ownership may arise either by assertion or by introducing a as the rival in a challenge. For a non-option token $\chi \in G^{\text{qry}} \cup W$, ownership arises by public disclosure in a justification. The live-set invariant is $a \in \Omega_t$ iff $\text{Own}_t(\text{Opt}(a)) \neq \emptyset$.

The state therefore has two sides: a dialogue side, which keeps track of public commitments and open burdens, and an explanatory side, which stores the causal information that has been made public so far.

Definition 4.2 (Relevance initialization). Assume a fixed policy $\text{InitRel}(i, a) \subseteq O \setminus \{a\}$ assigning to each agent i and option a an initial set of rivals. Whenever i first becomes a public owner of $\text{Opt}(a)$, the set $\text{Rel}_i(i, a)$ is initialized from this policy and may later be enlarged by challenges.

Relevance determines which pairwise comparisons an agent currently owes for a defended option. Because relevance ranges over all options, an option may be a relevant rival even when it is not currently live.

Definition 4.3 (Public deterministic SEM fragments). For each agent $i \in \text{Ag}$, $\widehat{M}_i^t = (G^{\text{exo}}, V_i^t, F_i^t, u^{\text{exo}})$, where G^{exo} is the fixed set of exogenous grounds, u^{exo} is their common public valuation, V_i^t is a finite set of endogenous Boolean variables, and F_i^t is a finite acyclic set of structural equations, with at most one equation for each endogenous variable. We require $V_i^t \subseteq G^{\text{qry}} \cup W \cup \{\text{Perm}_i^{a,b} \mid a, b \in O, a \neq b\}$.

The variable $\text{Perm}_i^{a,b} = 1$ means that, according to agent i 's current public fragment, a is publicly permissible relative to b . Each public SEM fragment is intentionally partial. It is not meant to represent the agent's full private theory, but only the endogenous variables and equations that have been explicitly disclosed.

Definition 4.4 (Equation replacement). Let F be a set of structural equations and let E be a finite set of equations with pairwise distinct left-hand-side variables. Write $F \oplus E$ for the equation set obtained from F by removing every equation in F whose left-hand-side variable occurs in some equation of E , and then adding the equations in E .

Definition 4.5 (Intervention on a public fragment). Let $X \subseteq G^{\text{qry}} \cup W \cup \{\text{Perm}_i^{a,b} \mid a, b \in O, a \neq b\}$ be a finite set of endogenous Boolean variables, and let $v : X \rightarrow \{0, 1\}$ be a valuation on X . The intervened fragment $\widehat{M}_i^t[X \leftarrow v]$ is obtained from $\widehat{M}_i^t$ by enlarging the endogenous-variable set with X and by replacing, for each $x \in X$, the structural equation for x if already present and otherwise adding the constant equation $x := v(x)$, leaving all other equations unchanged.

If $X = \{x_1, \dots, x_n\}$ and $v(x_k) = 0$ for all k , we also write $\widehat{M}_i^t[X \leftarrow \mathbf{0}]$. If $X = \{x_1, \dots, x_n\}$ and $v(x_k) = 1$ for all k , we also write $\widehat{M}_i^t[X \leftarrow \mathbf{1}]$. Because each F_i^t is acyclic and u^{exo} is fixed, every fragment $\widehat{M}_i^t$ has a unique solution. Intervention is used here to model direct public commitment effects of speech acts. In particular, assertion and challenge force the relevant permissibility variables to 1, while retraction forces them back to 0.

Definition 4.6 (Truth of targets in a public fragment). Write $\widehat{M}_i^t \models x = 1$ iff the unique solution of $\widehat{M}_i^t$ assigns value 1 to x . For a target set T whose members are formulas of the form $x = 1$, write $\widehat{M}_i^t \models T$ iff $\widehat{M}_i^t \models x = 1$ for every $x = 1 \in T$.

Outstanding targets are the semantic obligations generated by questions and challenges. They specify exactly what the next successful justification must make true in the updated public model.

Definition 4.7 (Outstanding targets). For $a \in O$, define the option-level target $Q_t(i, a) = \{\text{Perm}_i^{a,b} = 1 \mid b \in \text{Rel}_t(i, a)\}$. For a specific comparison of a with rival b , define the focal target $T_t(i, a, b) = \{\text{Perm}_i^{a,b} = 1\}$. For a public reason-element $\chi \in G^{\text{qry}} \cup W$, define the element target $E_t(i, \chi) = \{\chi = 1\}$.

The protocol distinguishes three kinds of justificatory burden: a broad burden for a defended option, a focal burden for a specific challenge, and a local burden for a previously disclosed reason-element.

Definition 4.8 (Package adequacy and admissible answers). Assume a fixed predicate $\text{Bal}(a, b, \Gamma)$ on finite non-empty sets Γ of contrastive reason elements. Read $\text{Bal}(a, b, \Gamma)$ as saying that the aggregate justifying weight in Γ for a against b is at least as strong as the aggregate requiring weight in Γ for b against a . We say that Γ is an admissible answer of agent i to target T at stage t , and write $\text{Ans}_t(i, T, \Gamma)$, iff

$$\begin{cases} \forall b \in \text{Rel}_t(i, a) \text{ Bal}(a, b, \Gamma_{a,b}), & \text{if } T = Q_t(i, a), \\ \text{Bal}(a, b, \Gamma), & \text{if } T = T_t(i, a, b), \\ \Gamma \subseteq G, & \text{if } T = E_t(i, \chi), \end{cases}$$

where $\Gamma_{a,b}$ is the subset of Γ containing the reason elements for a against b .

In order to keep the dialogue semantics neutral with respect to the underlying aggregation theory, we do not specify a particular formalism to be used for package adequacy.

Example 4.9 illustrates the difference between an explanation coming after a question and an explanation coming after a challenge. Example 4.10 shows that once a reason has been publicly disclosed, its elements can become the object of further questioning.

Example 4.9 (Option-level question and challenge). Let $\alpha, \beta \in O$. After an `assert`(A, α), `question`($B, A, \text{Opt}(\alpha)$) or `challenge`(B, A, α, β) open two different burdens. The question asks for a package Γ_q consisting of finite balanced subpackages for α against the currently relevant rivals. The challenge asks for a finite package Γ_c of contrastive reason elements for α against β such that the aggregate justifying weight in Γ_c is at least as strong as the aggregate requiring weight in Γ_c .

Example 4.10 (Element-level question). Let $\rho = \langle h, J(r, \alpha, \beta) = \text{high}, R(r, \alpha, \beta) = \text{low} \rangle$. After the public disclosure of ρ , the weight atom $J(r, \alpha, \beta) = \text{high}$ is publicly owned by the disclosing agent and may therefore be questioned. An answer has the form `justify`(A, Δ) where $\Delta \subseteq G$ is a finite set of grounds for that weight assignment.

Definition 4.11 (Public permissibility). Option a is publicly permissible for agent i at step t iff $\widehat{M}_i^t \models Q_t(i, a)$. Equivalently, for every $b \in \text{Rel}_t(i, a)$, $\widehat{M}_i^t \models \text{Perm}_i^{a,b} = 1$.

Public permissibility is therefore not stored as an extra primitive component of the state, but instead it is read off from the current public SEM fragment.

Definition 4.12 (Reason bodies, disclosed items, and induced equations). For a contrastive reason element $\rho = \langle g, J(r, a, b) = w_J, R(r, a, b) = w_R \rangle$, its SEM body is $\text{Body}(\rho) = g \wedge (J(r, a, b) = w_J) \wedge (R(r, a, b) = w_R)$. For a nonempty finite set Γ of contrastive reason

elements all for a against b , define $\text{Body}_{a,b}(\Gamma) = \bigwedge_{\rho \in \Gamma} \text{Body}(\rho)$. For a justification package Γ , define its question-worthy disclosure set by

$$\text{Disc}(\Gamma) = \begin{cases} \bigcup_{\rho \in \Gamma} (\text{elts}(\rho) \cap (G^{\text{qry}} \cup W)), & \text{if } \Gamma \text{ is a set of contrastive reason elements,} \\ \Gamma \cap G^{\text{qry}}, & \text{if } \Gamma \subseteq G. \end{cases}$$

If Γ answers target T of agent i , define the induced equation set $\text{Eq}_i^t(T, \Gamma)$ by

$$\text{Eq}_i^t(T, \Gamma) = \begin{cases} \{\text{Perm}_i^{a,b} := \text{Body}_{a,b}(\Gamma)\}, & \text{if } T = T_i(i, a, b), \\ \{\text{Perm}_i^{a,b} := \text{Body}_{a,b}(\Gamma_{a,b}) \mid b \in \text{Rel}_i(i, a)\}, & \text{if } T = Q_i(i, a), \\ \{\chi := \bigwedge_{g \in \Gamma} g\}, & \text{if } T = E_i(i, \chi) \text{ and } \Gamma \subseteq G, \end{cases}$$

where, in the second case, $\Gamma_{a,b}$ is the subset of Γ containing the reason elements for a against b .

At the SEM level, justification works by adding equations. A reason-element contributes a conjunctive body, and a package contributes the equations needed to make the relevant target true. Every atom occurring in Γ is publicly disclosed as true. If $g \in G^{\text{exo}}$, its truth is already fixed by u^{exo} . The important point is that justification does two things at once: it adds equations for the target under discussion, and it also makes the atoms occurring in the justificatory package publicly available for later questioning.

The transition rules now connect the speech acts of the protocol with concrete updates of the public state and the public SEM fragments.

Definition 4.13 (Public transition rules). Each move m determines a partial update map $\text{Upd} : s_t \rightarrow s_{t+1}$. For a given state s_t , $\text{Upd}(s_t)$ is defined only if the corresponding precondition holds in s_t . After the move, the postcondition determines s_{t+1} . All components not mentioned remain unchanged.

1. **Assertion.** For $m = \text{assert}(i, a)$:
Preconditions: $i \notin \text{Own}_t(\text{Opt}(a))$.
Postconditions: $\text{Own}_{t+1}(\text{Opt}(a)) = \text{Own}_t(\text{Opt}(a)) \cup \{i\}$, $\Omega_{t+1} = \Omega_t \cup \{a\}$,
 $\text{Rel}_{t+1}(i, a) = \text{InitRel}(i, a)$. Let $\text{PermVar}_i^{t+1}(a) = \{\text{Perm}_i^{a,b} \mid b \in \text{Rel}_{t+1}(i, a)\}$.
Then $\widehat{M}_i^{t+1} = \widehat{M}_i^t[\text{PermVar}_i^{t+1}(a) \leftarrow \mathbf{1}]$.
2. **Question.** For $m = \text{question}(j, i, x)$:
Preconditions: $j \neq i$, $i \in \text{Own}_t(x)$, $x \in \text{Tok}$.
Postconditions: $\mathcal{O}_{t+1} = \begin{cases} \mathcal{O}_t \cup \{Q_i(i, a)\}, & \text{if } x = \text{Opt}(a), \\ \mathcal{O}_t \cup \{E_i(i, \chi)\}, & \text{if } x = \chi \in G^{\text{qry}} \cup W. \end{cases}$
3. **Challenge.** For $m = \text{challenge}(j, i, a, b)$:
Preconditions: $j \neq i$, $a \neq b$, $i \in \text{Own}_t(\text{Opt}(a))$.
Postconditions: $\text{Own}_{t+1}(\text{Opt}(b)) = \text{Own}_t(\text{Opt}(b)) \cup \{j\}$, $\Omega_{t+1} = \Omega_t \cup \{b\}$,
 $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \{T_i(i, a, b)\}$, $\text{Rel}_{t+1}(i, a) = \text{Rel}_t(i, a) \cup \{b\}$.
If $j \notin \text{Own}_t(\text{Opt}(b))$, then $\text{Rel}_{t+1}(j, b) = \text{InitRel}(j, b) \cup \{a\}$;
otherwise $\text{Rel}_{t+1}(j, b) = \text{Rel}_t(j, b) \cup \{a\}$.
Let $\text{PermVar}_i^{t+1}(a, b) = \{\text{Perm}_i^{a,b}\}$, then $\widehat{M}_i^{t+1} = \widehat{M}_i^t[\text{PermVar}_i^{t+1}(a, b) \leftarrow \mathbf{1}]$.

Let $\text{PermVar}_j^{t+1}(b) = \begin{cases} \{\text{Perm}_j^{b,c} \mid c \in \text{Rel}_{t+1}(j,b)\}, & \text{if } j \notin \text{Own}_t(\text{Opt}(b)), \\ \{\text{Perm}_j^{b,a}\}, & \text{otherwise.} \end{cases}$

Then $\hat{M}_j^{t+1} = \hat{M}_j^t[\text{PermVar}_j^{t+1}(b) \leftarrow \mathbf{1}]$.

4. **Justification.** For $m = \text{justify}(i, \Gamma)$:

Preconditions: there exists a target $T \in \mathcal{O}_t$ such that T is an outstanding target of agent i and $\text{Ans}_t(i, T, \Gamma)$.

Postconditions: letting $U_i^t(T, \Gamma) = \text{Eq}_i^t(T, \Gamma) \cup \{\chi := 1 \mid \chi \in \text{Disc}(\Gamma)\}$, we have $V_i^{t+1} = V_i^t \cup \{x \mid x := \varphi \in U_i^t(T, \Gamma) \text{ for some } \varphi\}$, and $F_i^{t+1} = F_i^t \oplus U_i^t(T, \Gamma)$. For every $\chi \in \text{Disc}(\Gamma)$, $\text{Own}_{t+1}(\chi) = \text{Own}_t(\chi) \cup \{i\}$. Finally, $\mathcal{O}_{t+1} = \mathcal{O}_t \setminus \{T\}$. By construction, $\hat{M}_i^{t+1} \models T$.

5. **Retraction.** For $m = \text{retract}(i, u)$:

Preconditions: $i \in \text{Own}_t(u)$.

Postconditions:

- (a) If $u = \text{Opt}(a)$, let $\text{PermVar}_i^t(a) = \{\text{Perm}_i^{a,b} \in V_i^t \mid b \in \mathcal{O}, b \neq a\}$. Then $\text{Own}_{t+1}(\text{Opt}(a)) = \text{Own}_t(\text{Opt}(a)) \setminus \{i\}$, $\hat{M}_i^{t+1} = \hat{M}_i^t[\text{PermVar}_i^t(a) \leftarrow \mathbf{0}]$. $\mathcal{O}_{t+1} = \mathcal{O}_t \setminus \{T \in \mathcal{O}_t \mid T = Q_t(i, a) \text{ or } T = T_t(i, a, b) \text{ for some } b \in \mathcal{O}\}$. If $\text{Own}_{t+1}(\text{Opt}(a)) = \emptyset$, then $\Omega_{t+1} = \Omega_t \setminus \{a\}$.
- (b) If $u = \chi \in G^{\text{qy}} \cup W$, then $\text{Own}_{t+1}(\chi) = \text{Own}_t(\chi) \setminus \{i\}$, $\mathcal{O}_{t+1} = \mathcal{O}_t \setminus \{E_t(i, \chi)\}$, $\hat{M}_i^{t+1} = \hat{M}_i^t[\{\chi\} \leftarrow \mathbf{0}]$.

Taken together, these updates implement the intended cycle of the protocol: options are introduced, explanatory burdens are opened, justificatory equations are disclosed, and commitments can later be withdrawn again by intervention.

5. Case study

We extend the *Cafe or Kid* case introduced in Section 2 to a dialogue setting by having the protagonist accompanied by a friend, we will call them agents A and B . Instead of deciding on the option individually the two discuss whether they collectively should save the child (Save1), watch from the side as a bystander (Bystander) or continue to the cafe (Cafe). A suggests saving the child, to which agent B asks for justification. Agent A exclaims that the child is in danger, after which B proposes if it is not better to simply continue towards the cafe. Again A defends their position by the fact that the child is in danger, after which B agrees to withdraw the option of going to the cafe.

The set of options is then $\mathcal{O} = \{\text{Save1}, \text{Bystander}, \text{Cafe}\}$ and the grounds are $G^{\text{exo}} = \emptyset$ and $G^{\text{qy}} = \{g_{\text{danger}}\}$ representing that the child is in danger. The initially relevant options functions are $\text{InitRel}(A, \text{Save1}) = \{\text{Bystander}\}$ and $\text{InitRel}(B, \text{Cafe}) = \{\text{Save1}\}$. We start with an empty public state s_0 and will gradually expand it as the dialogue develops, $s_0 = \langle \Omega_0, \text{Own}_0, \text{Rel}_0, \mathcal{O}_0, (\hat{M}_A^0, \hat{M}_B^0) \rangle$ where $\Omega_0 = \mathcal{O}_0 = \text{Rel}_0(i, a) = \emptyset$ for all i and a , no agent owns any token and $\hat{M}_A^0 = (G^{\text{exo}}, V_A^0, F_A^0, u^{\text{exo}})$, $\hat{M}_B^0 = (G^{\text{exo}}, V_B^0, F_B^0, u^{\text{exo}})$ with $u^{\text{exo}} = V_A^0 = F_A^0 = V_B^0 = F_B^0 = \emptyset$.

Step 1. assert($A, \text{Save1}$)

State changes:

$\Omega_1 = \{\text{Save1}\}; \quad \text{Own}_1(\text{Opt}(\text{Save1})) = \{A\};$

$$\text{Rel}_1(A, \text{Save1}) = \{\text{Bystander}\}; \quad \hat{M}_A^1 = \hat{M}_A^0[\text{Perm}_A^{\text{Save1}, \text{Bystander}} \leftarrow \mathbf{1}].$$

Step 2. $\text{question}(B, A, \text{Opt}(\text{Save1}))$

State changes:

$$\mathcal{O}_2 = \{Q_t(A, \text{Save1})\}.$$

Step 3. $\text{justify}(A, \Gamma)$,

where $\Gamma = \{\langle g_{\text{danger}}, J(r_1, \text{Save1}, \text{Bystander}) = \text{high}, R(r_1, \text{Save1}, \text{Bystander}) = \text{low} \rangle\}$
and $\text{Bal}(\text{Save1}, \text{Bystander}, \Gamma_{\text{Save1}, \text{Bystander}})$.

State changes:

$$V_A^3 = V_A^2 \cup \{\text{Perm}_A^{\text{Save1}, \text{Bystander}}, g_{\text{danger}}, J(r_1, \text{Save1}, \text{Bystander}) = \text{high}, \\ R(r_1, \text{Save1}, \text{Bystander}) = \text{low}\};$$

$$F_A^3 = F_A^2 \oplus \{\text{Perm}_A^{\text{Save1}, \text{Bystander}} := g_{\text{danger}} \wedge J(r_1, \text{Save1}, \text{Bystander}) = \text{high} \wedge R(r_1, \text{Save1}, \text{Bystander}) = \text{low}, \\ g_{\text{danger}} := 1, (J(r_1, \text{Save1}, \text{Bystander}) = \text{high}) := 1, (R(r_1, \text{Save1}, \text{Bystander}) = \text{low}) := 1\}; \quad \text{Own}_3(g_{\text{danger}}) = \text{Own}_3(J(r_1, \text{Save1}, \text{Bystander}) = \text{high}) = \\ \text{Own}_3(R(r_1, \text{Save1}, \text{Bystander}) = \text{low}) = \{A\}; \quad \mathcal{O}_3 = \emptyset.$$

Step 4. $\text{challenge}(B, A, \text{Save1}, \text{Cafe})$

State changes:

$$\text{Own}_4(\text{Opt}(\text{Cafe})) = \text{Own}_3(\text{Opt}(\text{Cafe})) \cup \{B\}; \quad \Omega_4 = \Omega_3 \cup \{\text{Cafe}\};$$

$$\mathcal{O}_4 = \mathcal{O}_3 \cup \{T_t(A, \text{Save1}, \text{Cafe})\};$$

$$\text{Rel}_4(A, \text{Save1}) = \text{Rel}_3(A, \text{Save1}) \cup \{\text{Cafe}\}; \quad \text{Rel}_4(B, \text{Cafe}) = \text{InitRel}(B, \text{Cafe}) = \{\text{Save1}\}; \\ \hat{M}_A^4 = \hat{M}_A^3[\text{Perm}_A^{\text{Save1}, \text{Cafe}} \leftarrow \mathbf{1}]; \quad \hat{M}_B^4 = \hat{M}_B^3[\text{Perm}_B^{\text{Cafe}, \text{Save1}} \leftarrow \mathbf{1}].$$

Step 5. $\text{justify}(A, \Gamma)$,

where $\Gamma = \{\langle g_{\text{danger}}, J(r_2, \text{Save1}, \text{Cafe}) = \text{high}, R(r_2, \text{Save1}, \text{Cafe}) = \text{low} \rangle\}$ and
 $\text{Bal}(\text{Save1}, \text{Cafe}, \Gamma_{\text{Save1}, \text{Cafe}})$.

State changes:

$$V_A^5 = V_A^4 \cup \{\text{Perm}_A^{\text{Save1}, \text{Cafe}}, g_{\text{danger}}, J(r_2, \text{Save1}, \text{Cafe}) = \text{high}, R(r_2, \text{Save1}, \text{Cafe}) = \text{low}\}; \\ F_A^5 = F_A^4 \oplus \{\text{Perm}_A^{\text{Save1}, \text{Cafe}} := g_{\text{danger}} \wedge J(r_2, \text{Save1}, \text{Cafe}) = \text{high} \wedge R(r_2, \text{Save1}, \text{Cafe}) = \text{low}, \\ g_{\text{danger}} := 1, (J(r_2, \text{Save1}, \text{Cafe}) = \text{high}) := 1, (R(r_2, \text{Save1}, \text{Cafe}) = \text{low}) := 1\}; \\ \text{Own}_5(g_{\text{danger}}) = \text{Own}_5(J(r_2, \text{Save1}, \text{Cafe}) = \text{high}) = \text{Own}_5(R(r_2, \text{Save1}, \text{Cafe}) = \text{low}) = \\ \{A\}; \quad \mathcal{O}_5 = \emptyset.$$

Step 6. $\text{retract}(B, \text{Opt}(\text{Cafe}))$

State changes:

$$\text{Own}_6(\text{Opt}(\text{Cafe})) = \text{Own}_5(\text{Opt}(\text{Cafe})) \setminus \{B\};$$

$$\hat{M}_B^6 = \hat{M}_B^5[\text{Perm}_B^{\text{Cafe}, \text{Save1}} \leftarrow \mathbf{0}]; \quad \Omega_6 = \Omega_5 \setminus \{\text{Cafe}\} = \{\text{Save1}\}.$$

This dialogue illustrates three central features of Libra: assertions introduce live options, questions and challenges open different justificatory targets, and justifications explain public commitments by explicit SEM equations.

6. Related work

Libra belongs to the tradition of dialogue protocols for agent interaction. Formal studies of inter-agent dialogue and protocol design emphasized turn structure, dialogue type, and dialectical obligation, while Fatio isolated five locutions for argumentative interaction and equipped them with axiomatic and operational semantics [11,9,7,8]. Deliberation-oriented work then shifted attention from belief change to option choice [18]. Libra stays close to this tradition, but it replaces generic propositional content with option claims, contrastive reasons, and explicit weights.

Formal argumentation has been distinguished into three conceptualizations [22,23]: argumentation as inference, as dialogue, and as balancing. The first derives conclusions by constructing and evaluating arguments; the second studies rule-governed interaction between agents; and the third weighs pro and con considerations of alternatives. This typology locates Libra at the interface of dialogue and balancing. From the dialogue side, Libra inherits Fatio-style locutions and public justificatory burdens; from the balancing side, it inherits a contrastive treatment of reasons with explicit justifying and requiring weights. In this respect, it is close to recent work on balancing for agent decision making, which formalizes Tucker's dual-scale and tournament model by treating options as arguments and attacks as arising from pairwise comparisons over weighted reasons [20]. Libra differs by making this balancing process public and dynamic: agents can introduce options, enlarge relevance sets through challenges, disclose grounds and weights, and retract commitments. Its SEM layer then records how those disclosures determine pairwise permissibility variables, rather than treating the offered reasons as arguments alone.

7. Conclusion and further research

Libra models comparative deliberation through two connected layers: a public dialogue protocol and a public explanatory semantics. Assertions and challenges bring options into play; questions and challenges open burdens at option, rival, and element level; and justifications disclose grounds, weights, and equations in a speaker's public SEM fragment. This separates protocol from explanation while preserving the dynamic comparison-space perspective that motivates the framework. Several future directions follow naturally. One is theoretical: establish general properties of the update system, such as persistence, soundness of successful justifications, and complexity bounds for maintaining public permissibility under changing relevance sets. Second, on the computational and empirical side, Libra should be implemented within the LogiKey framework [2] and tested on richer ethical and legal cases; related LogiKey applications include [1,6,12,13,14]. A third direction connects Libra to explainable AI and agentic AI. Contrastive explanation asks why one outcome obtains rather than another [10], and recent philosophical work argues that reasons are often indexed to alternatives [17,19]. Libra provides a protocol-level setting for such explanations. In hybrid or agentic systems, large language models may propose options and justifications, while Libra constrains them through formal mechanisms, in line with the architecture in [16] and formally guided LLM explanations [21].

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